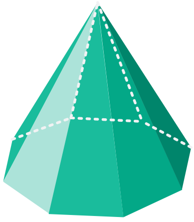
Solids

1 Name the following solids:

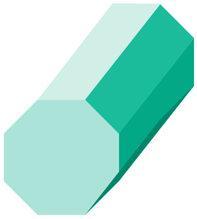
a

b

c



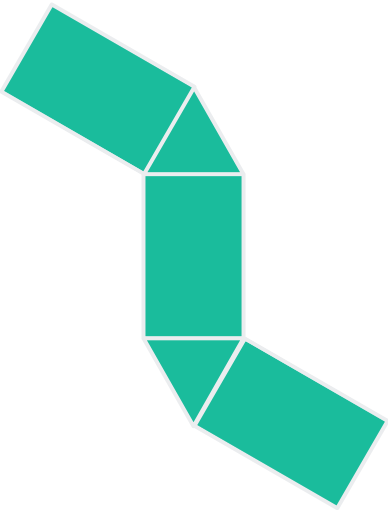
d



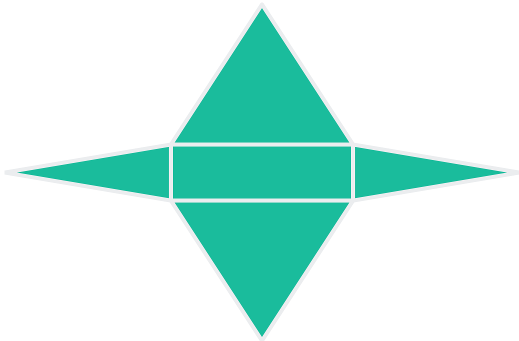
Nets

2 Name the solid formed from each of the following nets:

a

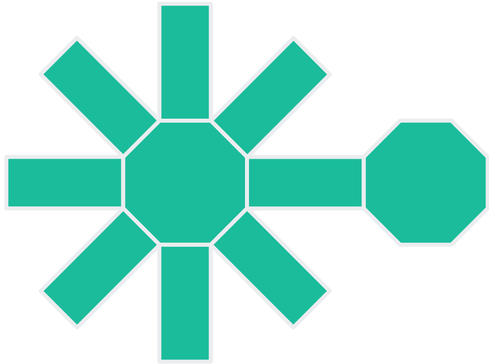


b



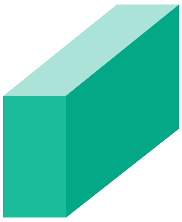
c

d



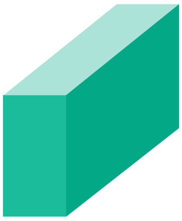
3 Draw a net for the following solids:

a

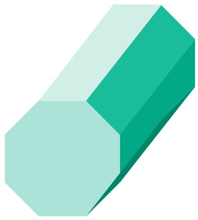


b

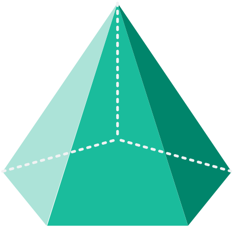
c



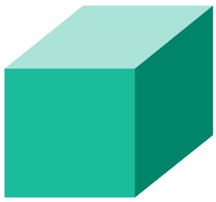
d



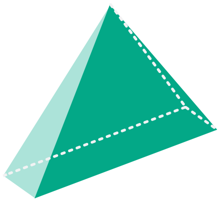
e



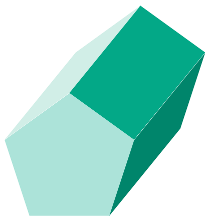
f



g



h

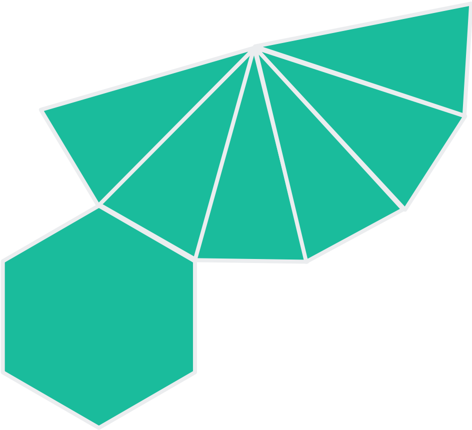


4 Name the 3D shapes formed by the following nets:

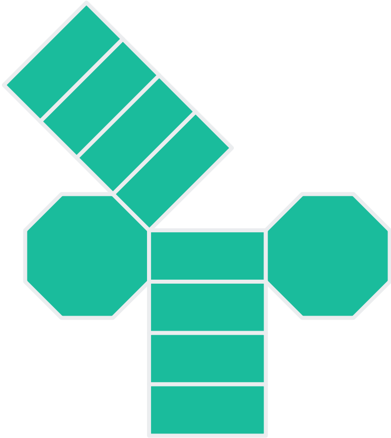
a

b

c

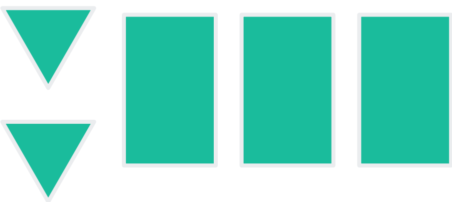


d

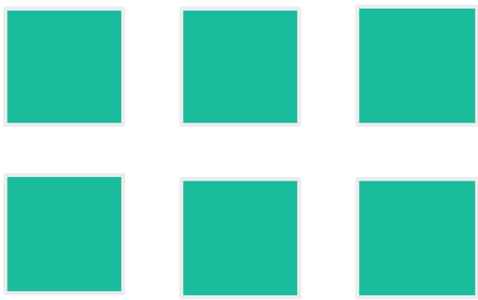


5 State the solid that can be formed from the given sets of 2D shapes:

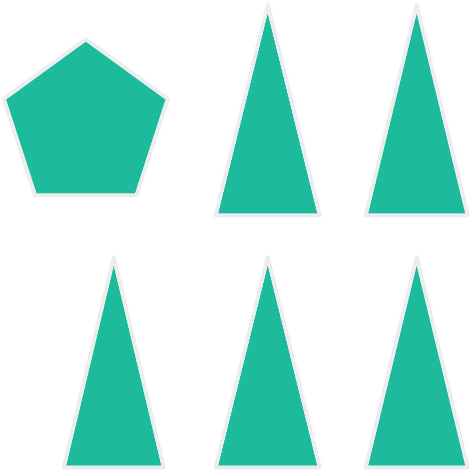
a



b



c



Front, side, and plan view

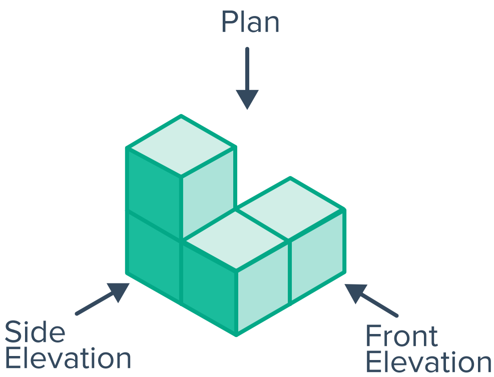
6 For each of the following solids formed from cubes:

i Draw the diagram for the side view.

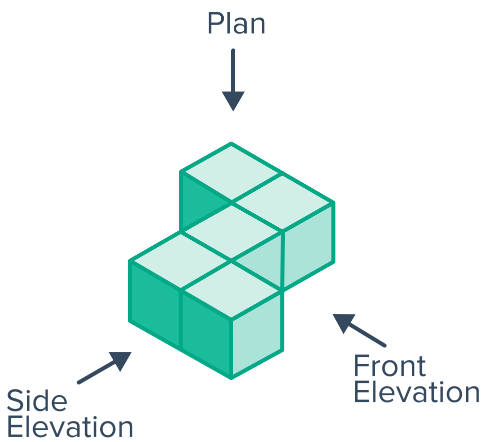
ii Draw the diagram for the front view.

iii Draw the diagram for the plan view.

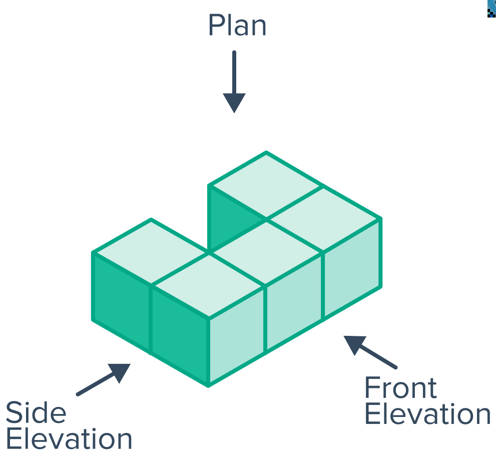
a



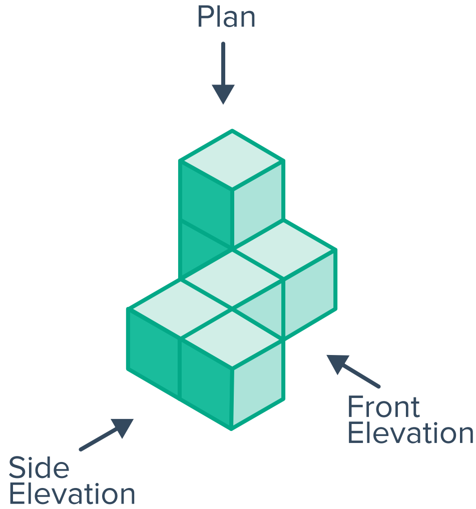
b



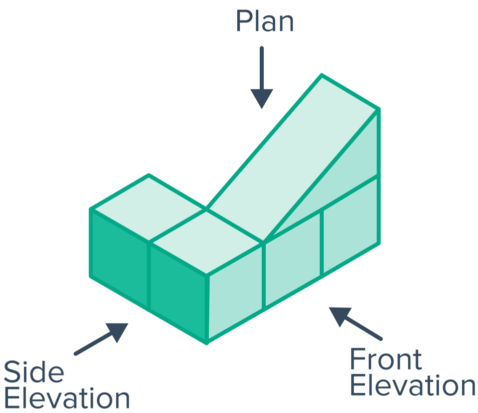
c


Plan
Side Elevation
Front Elevation

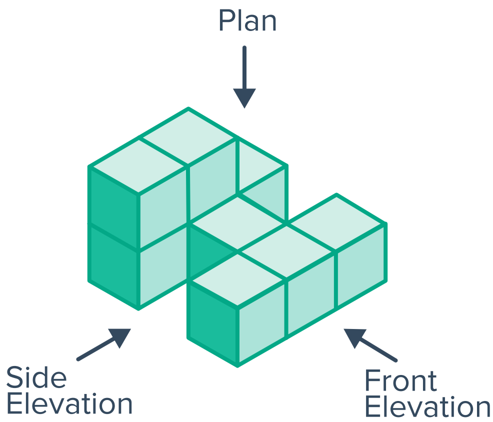
d


Plan
Side Elevation
Front Elevation

e

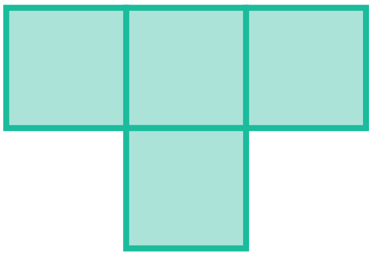

Plan
Side Elevation
Front Elevation

f

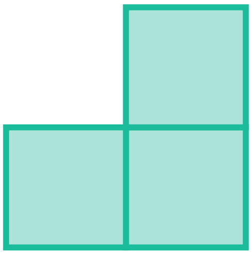

Plan
Side Elevation
Front Elevation

7 Select the solid that matches the given el[redacted]ns:

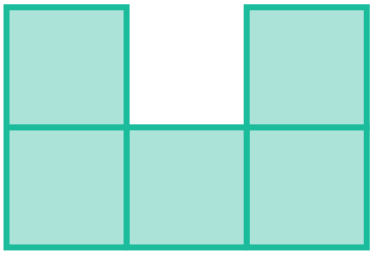
a



Plan

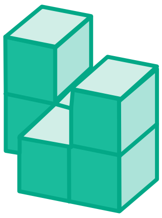


Front Elevation

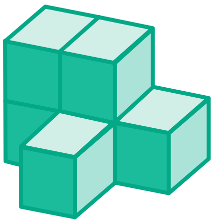


Side Elevation

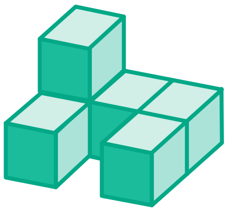
A



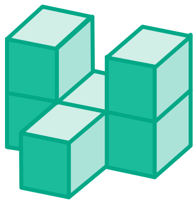
B



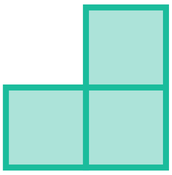
C



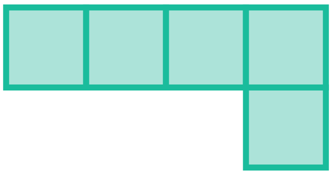
D



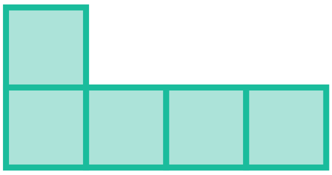
b



Front Elevation

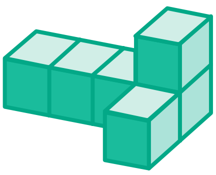


Plan

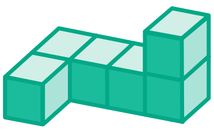


Side Elevation

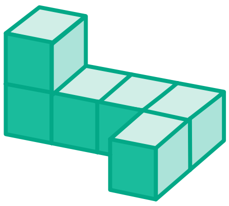
A



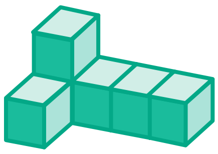
B



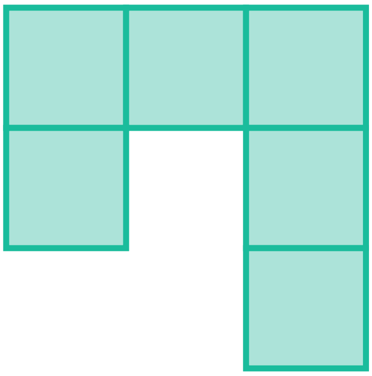
C



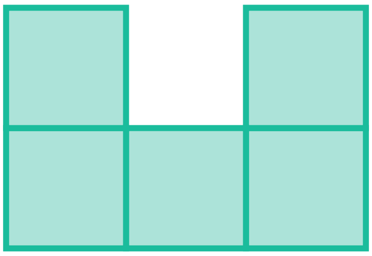
D



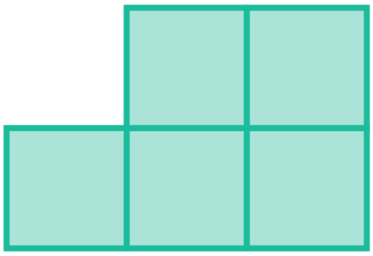
c



Plan

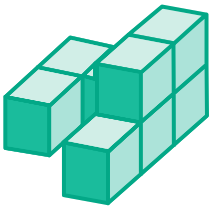


Side Elevation

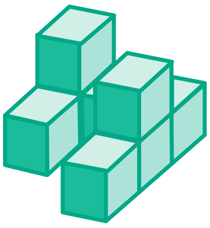


Front Elevation

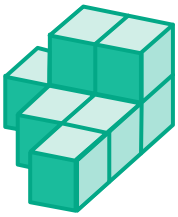
A



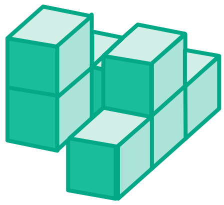
B



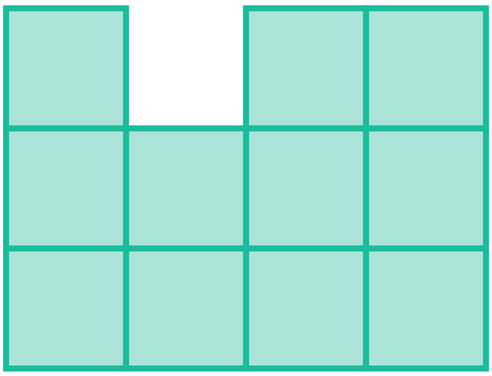
C



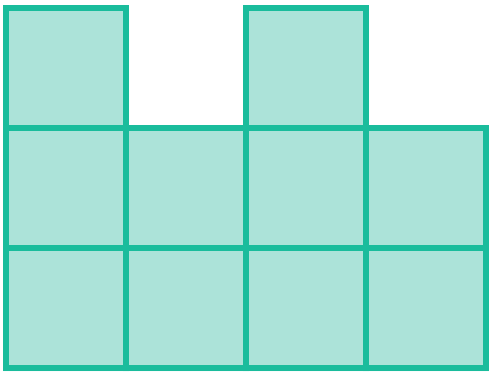
D



d

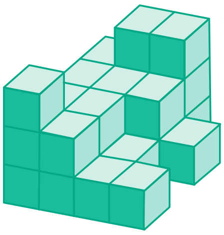


Front Elevation

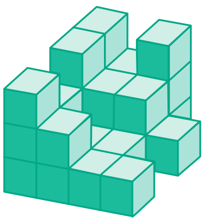


Side Elevation

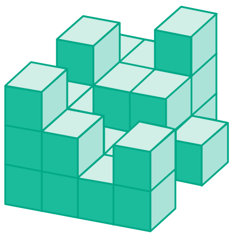
A



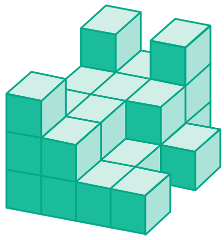
B



C



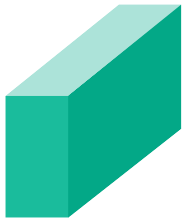
D



8 Name the following solids:

a

b



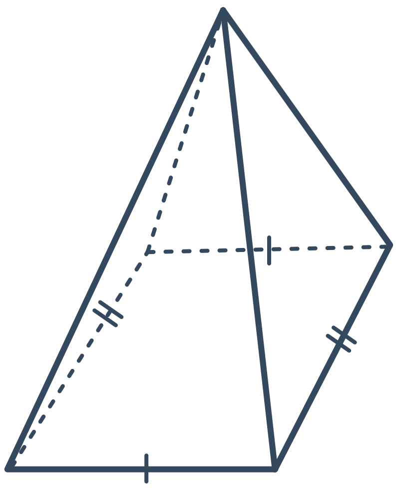
c

9 For each of the following solids:

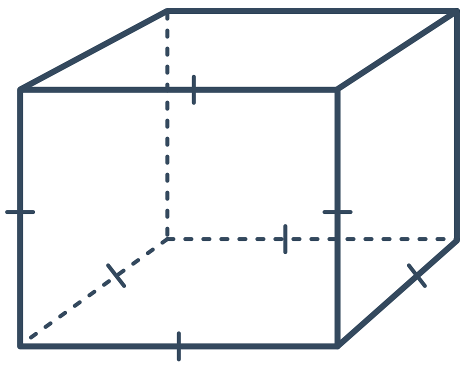
- **i** Name the solid.
- **ii** State whether the solid has a uniform cross-section.
- **iii** State the order of rotational symmetry about an vertical axis down the center of the solid.
- **iii** State whether the solid has plane symmetry.

a

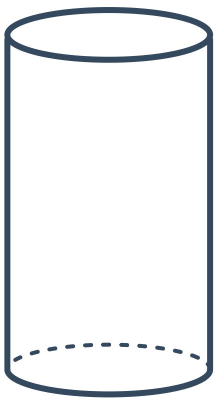
b



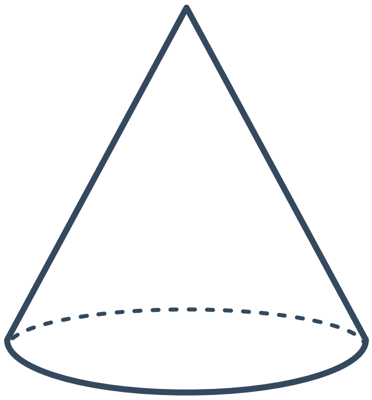
c



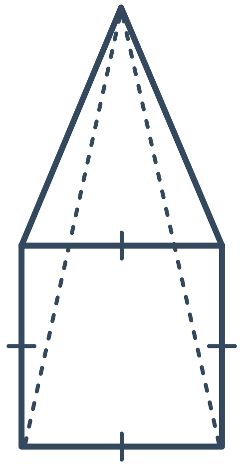
d



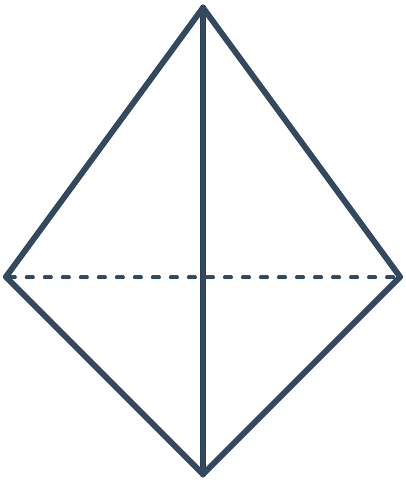
e



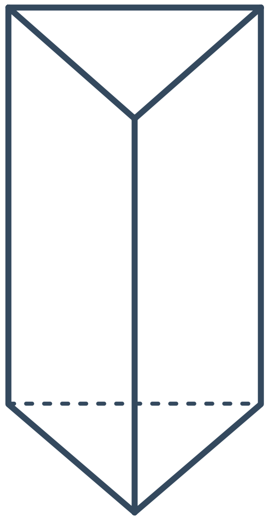
f



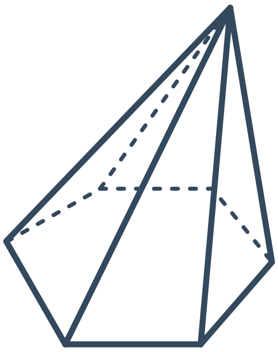
g



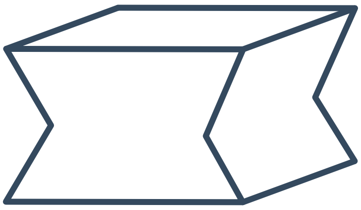
h



10 Explain why shapes A and B are polyhedra while shape C is not.



A

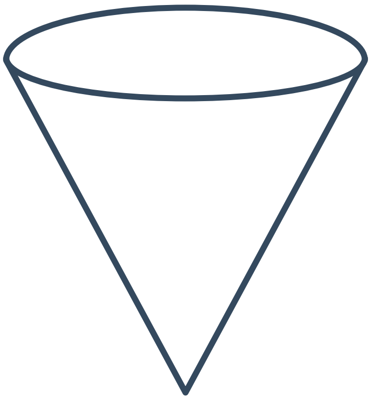


B

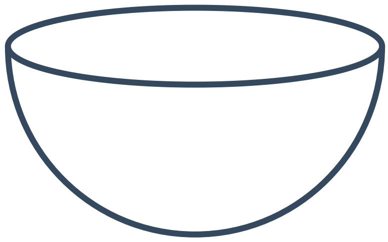
C

11 Determine whether the following solids ar[redacted]yhedrons:

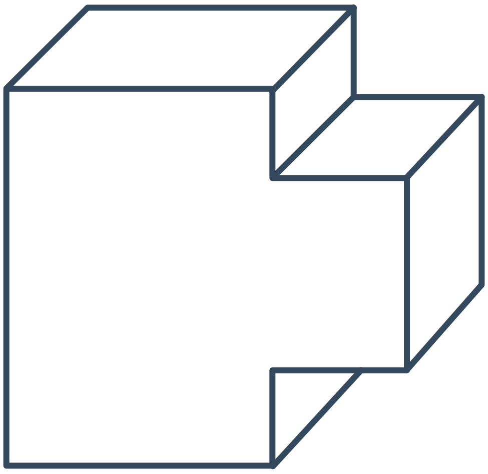
a



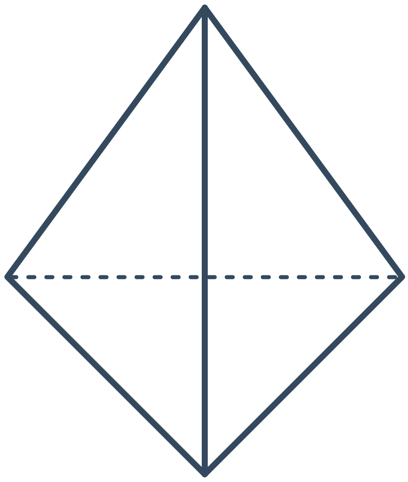
b



c



d



12 State whether the following solids have faces, vertices **and** edges:

a Sphere **b** Cone **c** Tetrahedron **d** Cylinder

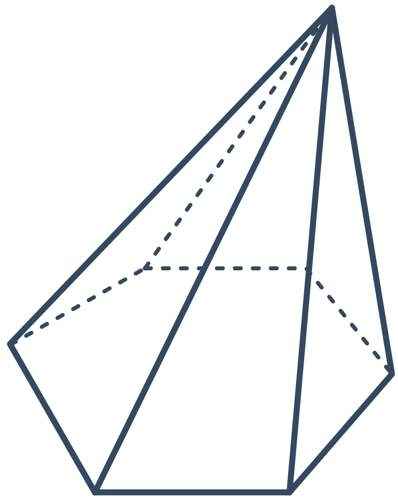
13 Consider a rectangular prism.

a How many faces meet at any one vertex?

b How many faces meet at any one edge?

c How many edges meet at any one vertex?

14 Consider the following hexagonal pyramid



a How many faces does it have?

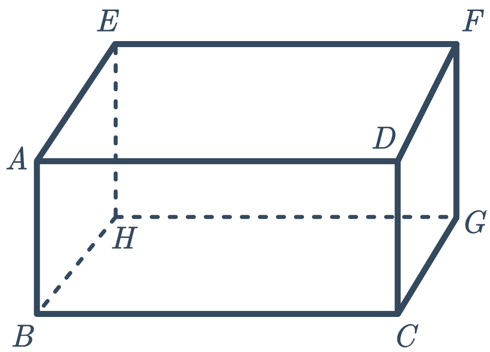
b How many vertices does it have?

c How many edges does it have?

15 Complete the table with the number of features for each solid:

3D shape	Number of faces	Number of vertices	Number of edges
Hexagonal prism			
Triangular pyramid			
Rectangular prism			

16 Consider given the rectangular prism:



a Name two parallel edges.

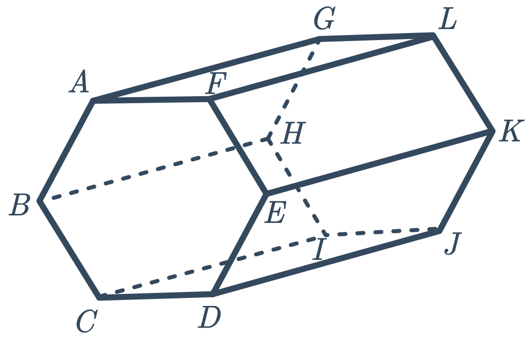
b Name two edges that are both intersecting and perpendicular.

c Name all of the edges that are parallel to BH.

d Name all of the edges that are perpendicular to BH.

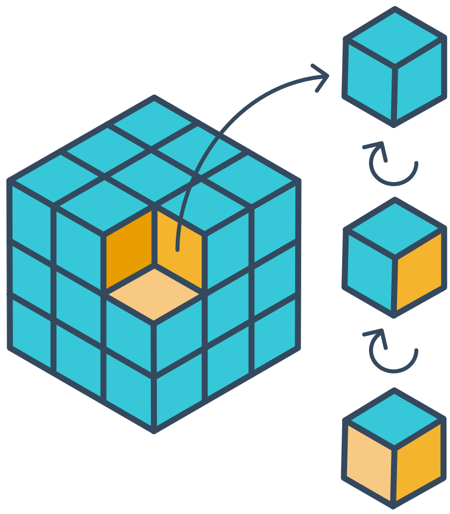
e Does the prism have plane symmetry about the plane $AFGB$?

17 Consider the given hexagonal prism:



- **a** Name the face that is parallel to $FEKL$.
- **b** How many faces are perpendicular to $ABCDEF$?
- **c** Name two parallel edges.
- **d** Name two perpendicular edges.
- **e** State whether the following are planes of symmetry for the prism:
  - **i** $BEKH$
  - **ii** $ADJG$
  - **iii** $BDJH$
  - **iv** $CFLI$

18 A number of $1 \text{ cm} \times 1 \text{ cm} \times 1 \text{ cm}$ cubes are glued together to form a $3 \text{ cm} \times 3 \text{ cm} \times 3 \text{ cm}$ cube as shown:



The outside faces of the large cube are painted blue and the small cubes are then pulled apart.

- **a** How many of the small cubes will have at least one face painted blue?
- **b** How many of the small cubes will have at least two faces painted blue?
- **c** How many of the small cubes will have three faces painted blue?

19 Euler related the number of faces, edges a[redacted]ertices of any polyhedron in one formula.

a Complete:

Solid	Number of Faces (F)	Number of Vertices (V)	Number of Edges (E)	$F+V$
Rectangular Prism		8		
Triangular Prism		6		
Hexagonal Prism		12		
Triangular Pyramid		4		
Pentagonal Pyramid		6		

b Determine whether each of the following rules correctly states the relationship between the number of faces, vertices and edges of a polyhedron:

i $F+V-E=2$

ii $F+V+E=2$

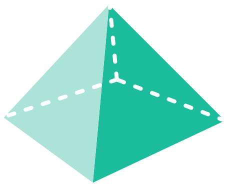
iii $E+V-F=2$

iv $F+E-V=2$

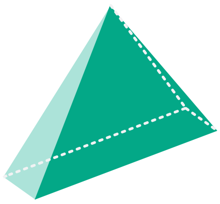
20 Select the right name for each solid from the list below:

- Square pyramid
- Rectangular pyramid
- Triangular prism
- Rectangular prism
- Octagonal pyramid
- Octagonal prism

a



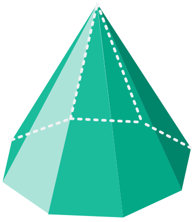
b



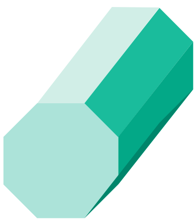
c

d

e



f



21 How many planes of symmetry do the following solids have?

a Cube

b Sphere

c Cone

d Square prism